Context-Aware Safety Monitor

Aegis Sentinel: Next-Generation Real-Time Vision & Behavior Analytics

Executive Summary

The **Context-Aware Safety Monitor (Aegis Sentinel)** is an enterprise-grade, edge-AI computer vision and spatial intelligence system designed to autonomously monitor workplace environments, industrial facilities, and public infrastructure. By combining multi-object tracking, MediaPipe 3D pose estimation, and a specialized PyTorch LSTM sequential action neural network, Aegis Sentinel transforms standard CCTV camera networks into proactive safety monitors.

Unlike legacy CCTV systems that require continuous manual human monitoring, Aegis Sentinel continuously evaluates spatial proximity, posture aspect ratios, hot-zone boundaries, and rapid kinetic trajectories to detect critical threats (such as worker falls, physical aggression, and restricted zone breaches) within milliseconds.

Status Update: Operational Readiness Confirmed
All identified backend dependency conflicts, DeepSORT tracker fallback logic, LSTM synthetic training indexing, Windows webcam capture synchronization, stream WebSocket concurrency, SQLite schema migrations, and frontend IP networking issues have been 100% resolved and verified via automated build pipelines.

Core System Capabilities & Architecture

Module	Technology Stack	Function & Capabilities
Person Detection	YOLOv8 (PyTorch / Ultralytics)	High-speed, real-time person bounding box detection operating on RTSP/MP4/Webcam streams.
Multi-Object Tracking	DeepSORT + Centroid/IOU Fallback	Maintains persistent subject identity tracking (IDs) across video frames with seamless fallback if DeepSORT weights fail.
3D Pose Estimation	MediaPipe Pose Analytics	Extracts 12 key joint spatial landmarks (shoulders, elbows, wrists, hips, knees, ankles) relative to subject bounding box.
Behavior Analysis	PyTorch LSTM Action Network	Evaluates 30-frame temporal keypoint sequences to classify Normal behavior, Person Falls, and Aggression/Violence.
Geofencing & Hot Zones	Normalized Bounding Box Spatial Engine	Supports custom active monitoring zone calibration (min/max X & Y coordinates) to focus compute on high-risk areas.
Live HUD & Streaming	FastAPI, WebSockets, React & Vite	Streams low-latency JPEG overlays with telemetry indicators, audio alerts, and real-time threat index reporting.

Real-World Industrial & Commercial Utility

Aegis Sentinel is built to address critical safety challenges across diverse real-world environments:

1. Manufacturing & Heavy Industrial Facilities

- **Machine Danger Zones:** Prevents accidental entry into robotic work cells or conveyor lanes by triggering instant audio alarms.

- **Proximity Detection:** Measures relative distances between workers to prevent crushing hazards or tight-space collisions.

2. Warehouses & Logistics Hubs

- **Forklift & Pedestrian Safety:** Monitors forklift pathways and alert operators when human workers move dangerously close.
- **Slip, Trip & Fall Auditing:** Captures immediate photographic snapshots and logs database events whenever a collapse occurs.

3. Healthcare, Assisted Living & Hospitals

- **Autonomous Patient Fall Alerts:** Immediately notifies nursing staff when a patient falls in a hallway or room without requiring wearable sensors.
- **Privacy-Respecting Monitoring:** Analyzes numerical skeletal keypoints rather than storing raw unencrypted facial footage.

4. Smart Buildings & Public Infrastructure

- **Violence & Aggression Detection:** Identifies sudden high-frequency erratic movement patterns indicative of altercations.
- **Perimeter Geofencing:** Monitors building entry points and restricts access to designated zones outside operating hours.

Resolved Errors & Operational Verification

During the comprehensive codebase review, 8 technical issues were identified and fully resolved:

Component	Root Cause / Issue Description	Resolution Strategy
Dependencies	OpenCV binary conflict & missing packages.	Cleaned requirements.txt, removed opencv-python-headless, added deep-sort-realtime and updated Pillow.
Tracker Engine	DeepSORT frame crash on empty detections.	Added empty detection track processing and fallback to SimpleFallbackTracker.
Action LSTM	Keypoint array indexing & device mismatch.	Corrected left/right joint stride indexing, aligned CPU/GPU device handling, and trained fresh model weights.
Camera Capture	Threading race condition & Windows webcam load.	Added threading.Lock around frame buffer reads/writes and implemented cv2.CAP_DSHOW fallback.
Stream Management	Duplicate pipelines & orphan WS tasks.	Implemented async stream_lock and stop_stream(camera_id) invalidation upon camera edit/deletion.
Database	Blind ALTER TABLE on existing SQLite columns.	Inspected schema using PRAGMA table_info(cameras) before executing conditional migration ALTER statements.
Frontend Net	Hardcoded http://localhost:8000 URLs.	Centralized dynamic hostname resolution (window.location.hostname) for HTTP REST and WebSockets.

System Execution & Deployment Guide

1. Start Python FastAPI Backend:

```
uvicorn backend.main:app --host 0.0.0.0 --port 8000
```

2. Start React Dashboard Frontend:

```
cd frontend && npm run dev
```

3. Access Operations Dashboard:

Open your web browser at `http://localhost:5173` (or host IP address on local network).